Chapter 1 Start developing App with MIT App Inventor 2

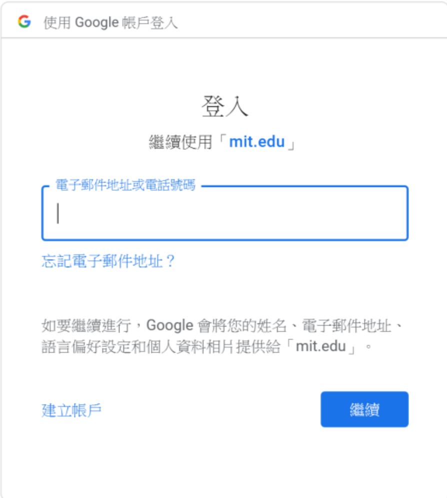
Key Words

Online Platform 網上平台	Component 元件 / App 中的物件	Visible 可視的
Non-visible 不可視的	Interface 界面 / 程式畫面	

1.1 Create a project in online platform[1]: App Inventor 2

Notes

- Step 1: Open URL: http://ai2.appinventor.mit.edu/ in Google Chrome.



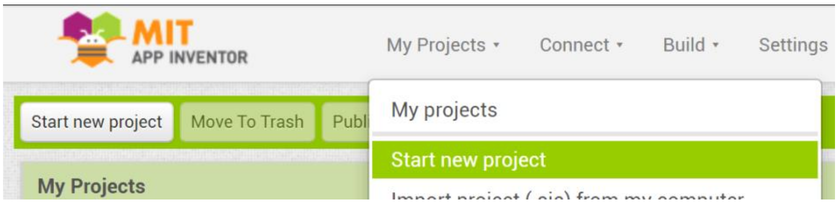
- Step 2: Login with Google (Use your school Google account).

- Step 3: Add a new project:

  Select "My Projects" → "Start new project"

  or

  Click the button "Start new project"



*** You should enter a name,

(i) begin with a letter,

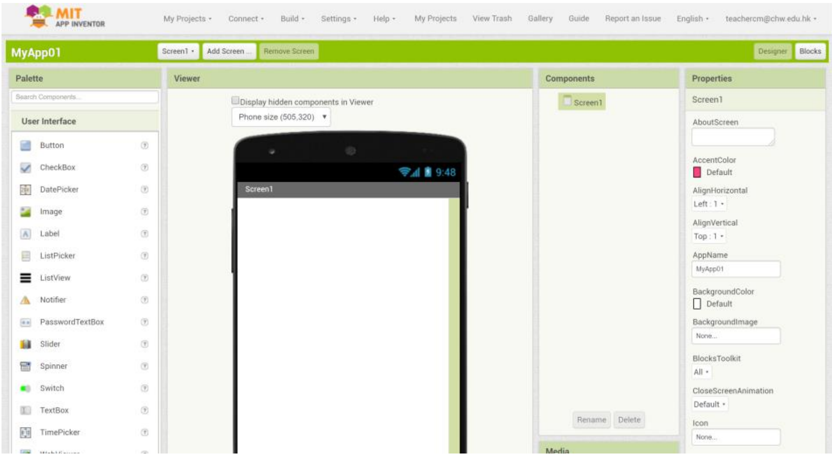
(ii) contain no spaces,

(iii) only contain letters, numbers and underscores.

[1] Online Platform 網上平台

Notes

- Step 4: You may now start developing your first App.



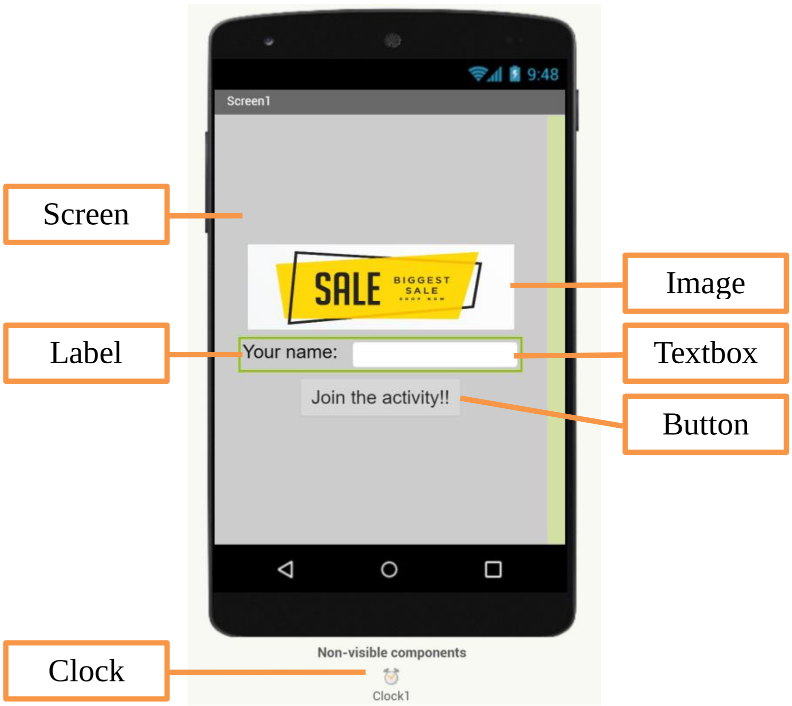
1.2 Introduction to five common **components**[1]

Examples of **visible**[2] component:

- *User Interface* → *Buttons* – It is used to detect click.
- *User Interface* → *Image* – It is used to display image/picture.
- *User Interface* → *Label* – It is used to show text.
- *User Interface* → *TextBox* – It is used for user text input.

Example of non-visible[3] component:

- *Sensors* → *Clock* – It is used to fire a timer to perform specific task.



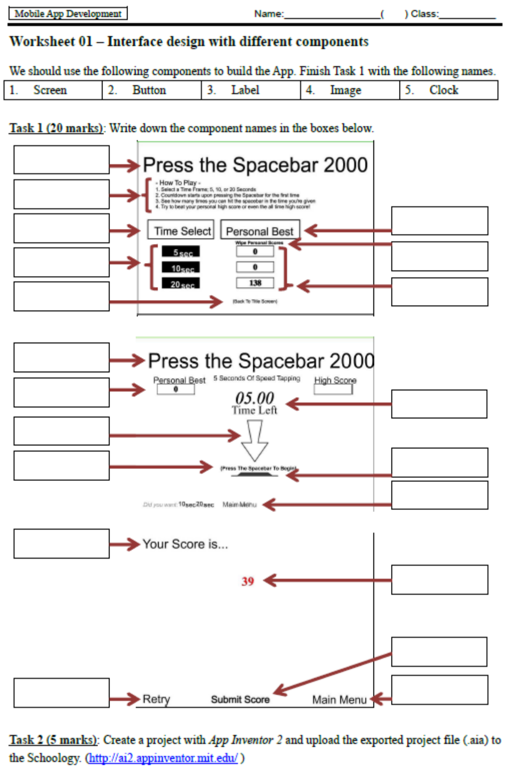
Example of an user interface

[1] Component 元件 / App 中的物件
[2] Visible 可視的
[3] Non-visible 不可視的

1.3 Homework

Task 1:

- Finish "Worksheet 01". Name the component in the **interfaces**[1] given.
- Scan you homework into PDF format and submit to the Schoology.



Mobile App Development Name:______________( ) Class:_________

Worksheet 01 – Interface design with different components

We should use the following components to build the App. Finish Task 1 with the following names.

1. Screen	2. Button	3. Label	4. Image	5. Clock

Task 1 (20 marks): Write down the component names in the boxes below.

Task 2 (5 marks): Create a project with *App Inventor 2* and upload the exported project file (.aia) to the Schoology. (http://ai2.appinventor.mit.edu/)

Worksheet 01

Task 2:

- Login MIT App Inventor 2 (http://ai2.appinventor.mit.edu).
- Start a new project.
- Add some components on Screen1. E.g. Image, Label, Textbox, Button, …
- Export the project into (.aia) format and submit to the Schoology.

[1] Interface 界面 / 程式畫面

Chapter 2　Steps of designing interface of an App

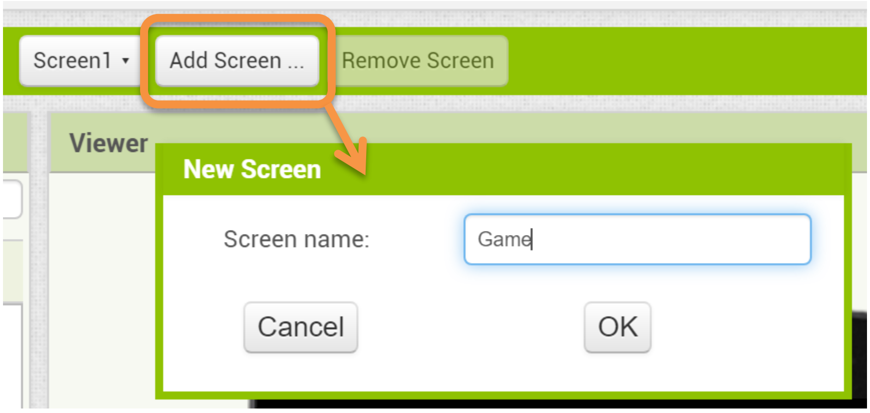
Key Words

Default 預設	Layout 佈局 / 配置 / 設計	Property / Properties 屬性
Scrollable 可捲動	Grid 格 / 格子	

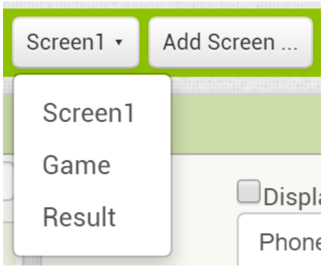
Notes

2.1 First step of developing App: Add Screen

- By default[1], the screen "Screen1" should be the startup screen.
- You should design the structure of your App by adding the screens needed.
- For example, adding a screen "Game".



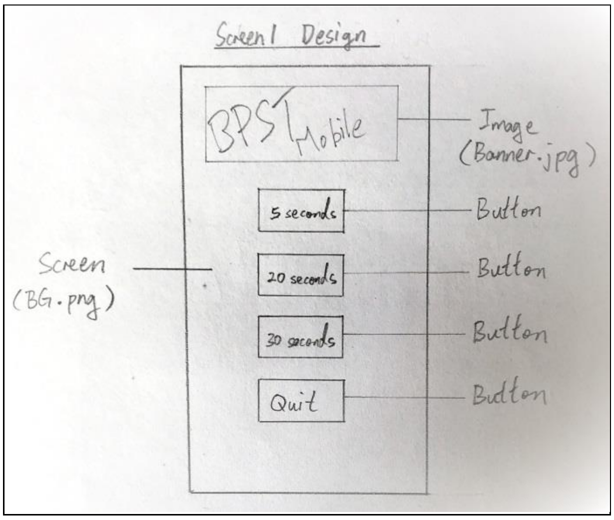
- After adding two screens, "Game" and "Result", the following figure shows an App with 3 screens.



[1] Default 預設

2.2 Second step of developing App: Interface design

- You should design the interface of every screen.



An example of the Main Menu interface design

- When designing an interface, you should consider the followings:
  1. **Layout**[1] of the interface;
  2. Components needed on the screen;
  3. **Properties**[2] of every component.

Layout – *The way in which text or pictures are set out on a screen.*

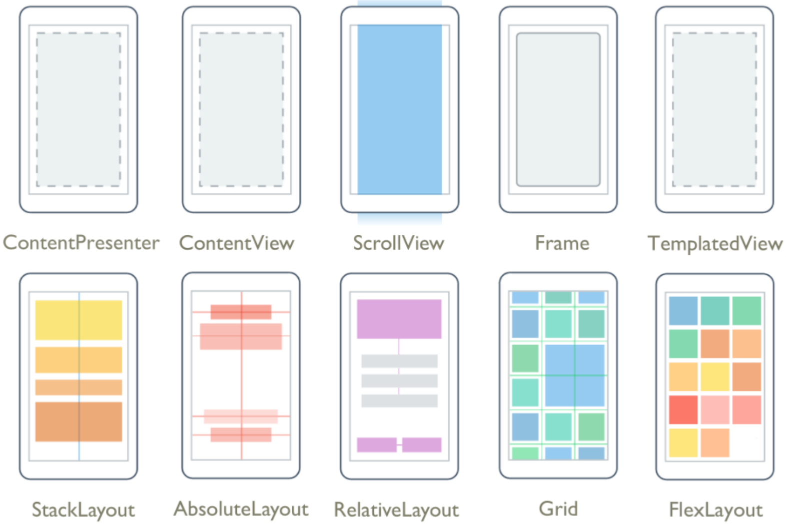


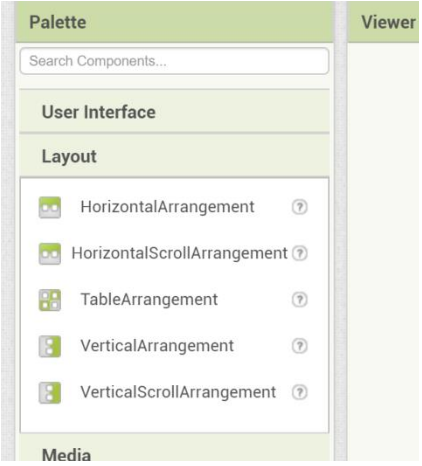
Image source:
https://docs.microsoft.com/zh-tw/xamarin/xamarin-forms/user-interface/controls/layouts

Notes

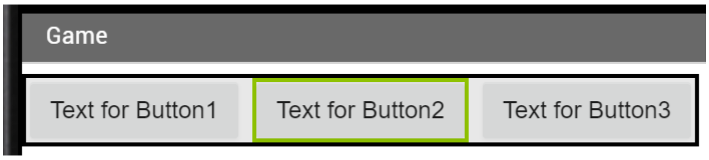
[1] Layout 佈局 / 配置 / 設計
[2] Property / Properties 屬性

Notes

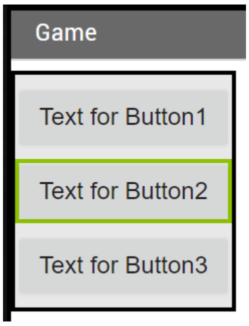
- In App Inventor 2, there are several types of layout component.



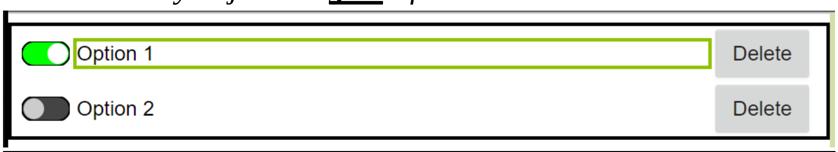
1. *Layout → HorizontalArrangement* or
   Layout → HorizontalScrollArrangement
   - *Arrange objects from left to right (non-scrollable or* ***scrollable***[1]*)*



2. *Layout → VerticalArrangement* or
   Layout → VerticalScrollArrangement
   - *Arrange objects from top to bottom (non-scrollable or scrollable)*



3. *Layout → TableArrangement*
   - Array objects in ***grid***[2] *form*



[1] Scrollable 可捲動
[2] Grid 格 / 格子

Components – *the objects in an App.*

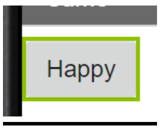
(Reference: http://ai2.appinventor.mit.edu/reference/components/)

- The following table shows some components in the categories "*User Interface*" and "*Layout*". (You can refer the reference site above to see the full list of components and their functions)

User Inteface	*Layout*
Button	HorizontalArrangement
CheckBox	HorizontalScrollArrangement
Image	TableArrangement
Label	VerticalArrangement
PasswordTextBox	VerticalScrollArrangement
Screen	
Switch	
TextBox	

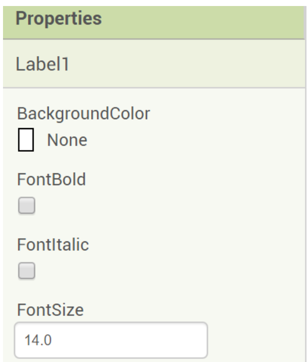
Properties – *The values of the properties represent the state of the component.*

- E.g. Button.Text = "Happy"
  It means that the text "Happy" shows on the button.



- E.g. Label1.BackgroundColor = Red
  It means that set the background color of Label1 to red.

- E.g. Label1.FontSize = 20
  It means that set the font size of text on Label1 to point 20.

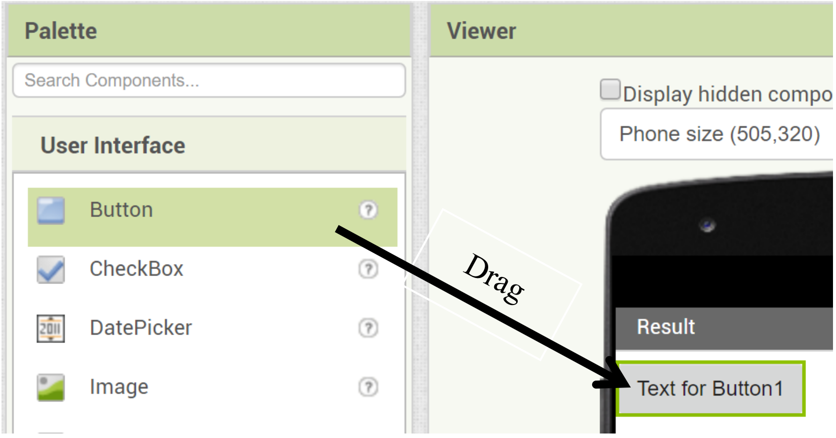


Notes

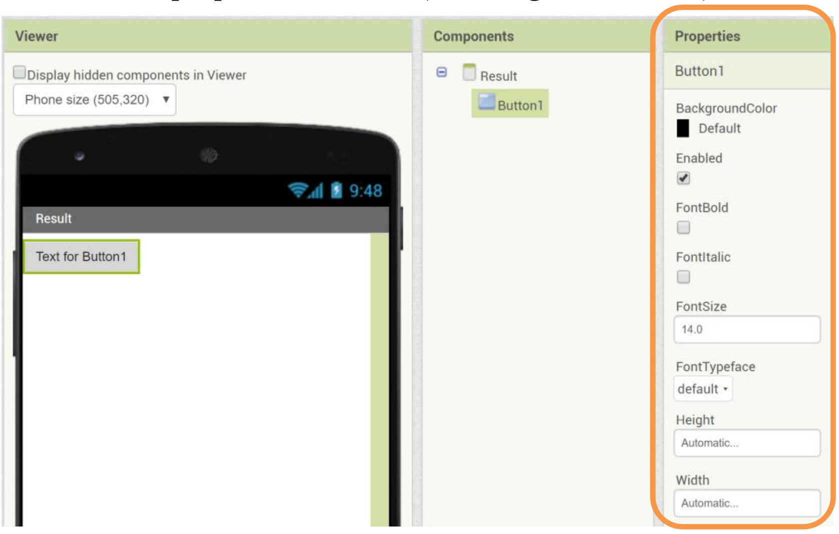
Add component to the Screen and change its properties.

Notes

- Add components into every screen by dragging components into screens.



- Select the component and change its properties by changing the values in the properties window (at the right-most area).



- The following is the introduction of some common properties.
    1. *BackgroundColor* – It indicates the color of the component.
    2. *Enabled* – It indicates user can use this component or not.
    3. *FontSize* – It indicates the size of text on the component.
    4. *Height* – It indicates the height of the component.
    5. *Width* – It indicates the width of the component.
    6. *Text* – It indicates the text shown on the component.
    7. *Visible* – It indicates the component can be seen or not.

- You can find all properties introduction of components in the web-site (http://ai2.appinventor.mit.edu/reference/components/).

2.3 Homework

Task 1 (10 marks):

You are a mobile App developer. Before program the App "Button Pressing Speedtest (BPST Mobile)" in App Inventor 2, you should design 3 interfaces (Screen1, Game, Result) by

1. Sketching the layout and components on each screen.
2. Name the components on each screen.

You should design the interfaces "Game" and "Result" by drawing.

Task 2 (5 marks):

In MIT App Inventor 2, finish the following steps:

Step 1: Start a new project named "BPST_3201". (Please use your own class and class number)

Step 2: Use "Add Screen…" function to add 2 screens named "Game" and "Result".

Step 3: In every screen, try adding components according to your design in Task 1 and design the interface "Screen1" as well.

Step 4: Try to edit the "Properties" of every components.

Step 5: Export your project to (.aia), and submit to the Schoology.

Chapter 3 Add programming code (blocks) in an App

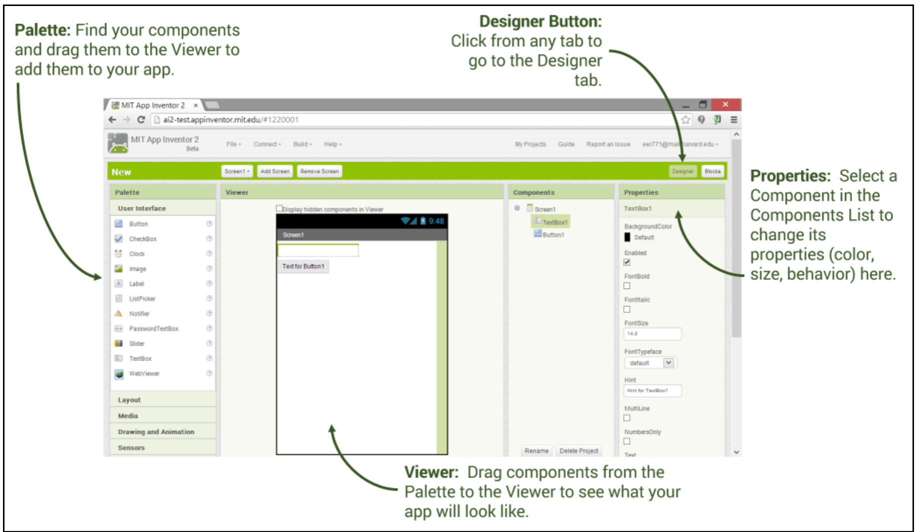
Key Words

Designer mode 界面設計模式	Blocks mode 編程模式	Behaviour 行為表現
Event （觸發）事件	Program Log 程式日誌	

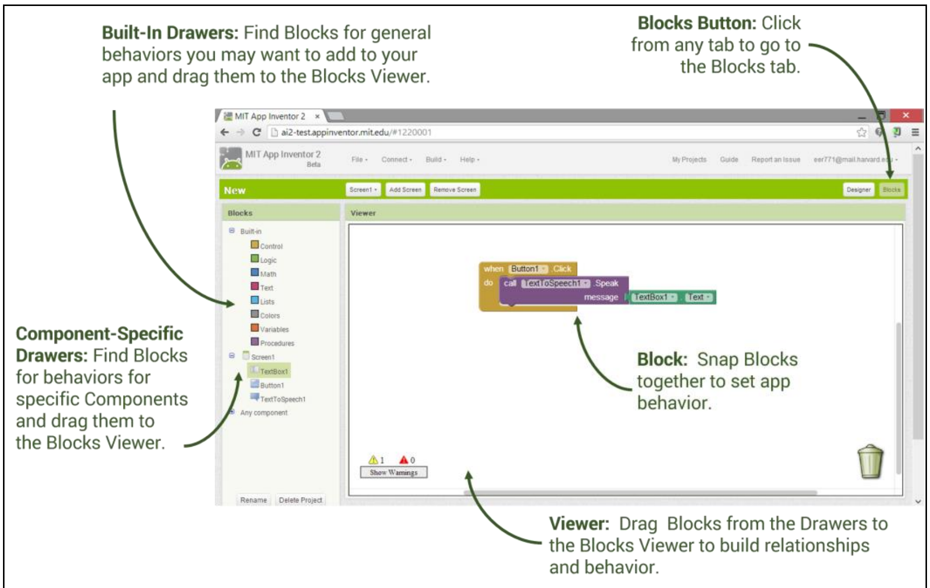
3.1 Third step of developing App: Add programming code (Blocks)

Notes

- There are two modes in MIT App Inventor 2: "**Designer mode**[1]" and "**Blocks Mode**[2]".



Designer Mode



Blocks Mode

[1] Designer mode 界面設計模式
[2] Blocks mode 編程模式

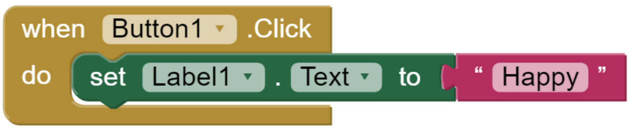
Notes

- There are two buttons "Designer" and "Blocks" at the top-right hand corner. They are used to switch between "Designer Mode" and "Blocks Mode" respectively.

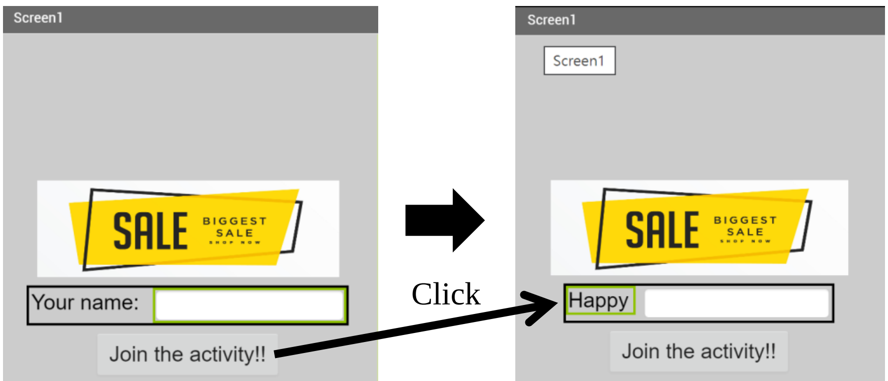
- **Designer Mode** – In this mode, you can:
  1. Add screens;
  2. Set the layout of the interface;
  3. Add components;
  4. Change the properties.

- **Blocks Mode** – In this mode, you can add the programming code (Blocks) to control the **behaviour**[1] of the App.
  For example:

  1. When the component "Button1" is clicked,
     the text of the "Label1" will change to "Happy".



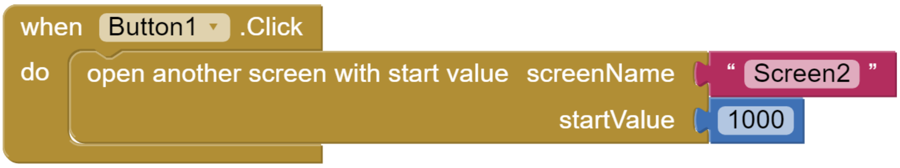
The following figure shows the effect of the above blocks.



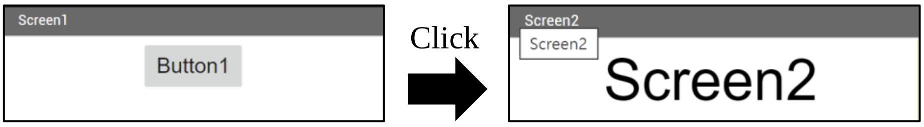
[1] Behaviour 行為表現

Notes

2. When the component "Button1" is clicked, another screen named "Screen2" will be opened with a starting value 1000.

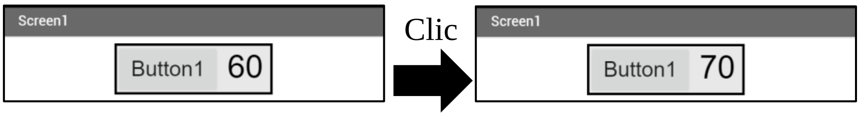


The following figure shows the effect of the above blocks.



3. When the component "Button1" is clicked, the text of "Label1" will be increased by 10.

The following figure shows the effect of the above blocks.



○ Besides the **event**[1] "Click" of a button, there are many other events of different components. Please check the following reference links to learn more.

http://ai2.appinventor.mit.edu/reference/blocks/

http://ai2.appinventor.mit.edu/reference/components/index.html

[1] Event （觸發）事件

3.2 Homework

Task 1 (5 marks): - Adding blocks.
In MIT App Inventor 2, finish the following steps:

Step 1: Open the project "BPST_2E01", which was built in last week.

Step 2: Make sure that you have finished all interfaces design (Designer Mode).

Step 3: Add blocks to do the following (Blocks Mode).

- To control the screen changing.
- To update the count shown on the Label when the user clicks on the button.

Step 4: Export your project to a file (.aia) and submit to the Schoology.

Task 2 (5 marks): - Keeping **program log**[1].
Please copy the programming code from your project. You should keep the log in a systematic way. Therefore, in the spaces provided below, please write down the blocks that you have used in different screens. An example in "Screen1" is shown below.

Screen "Screen1"

```
When Button1.Click
do  open another screen with start value    screenName    "Game"
                                             startValue       0
```

Please keep your program log in "Worksheet 03".

[1] Program Log 程式日誌

Chapter 4 Add programming code (blocks) in an App

Key Words

Fire 觸發	Interval 間距	Non-visible 不可視的
Countdown 倒數	Program Log 程式日誌	

4.1 Introduction to "Sensors → Clock"

Notes

- It provides the system date and time using the internal clock on the phone.
- It can **fire**[1] a timer at regularly set **intervals**[2].

An example of using "Sensors → Clock" – A timer program.

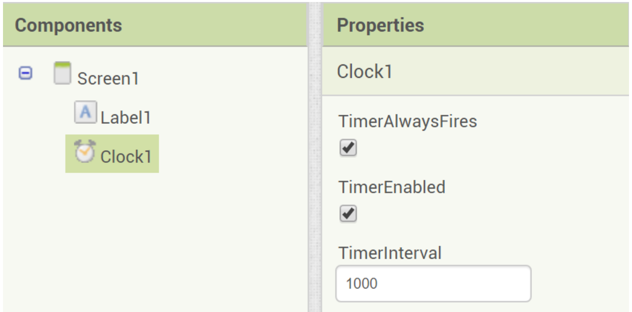
Step 1: Add the component "Sensors → Clock"

- You can add "Sensors → Clock" to the screen by dragging the clock to the screen.
- The clock will be added as a **non-visible**[3] component.

[1] Fire 觸發
[2] Interval 間距
[3] Non-visible 不可視的

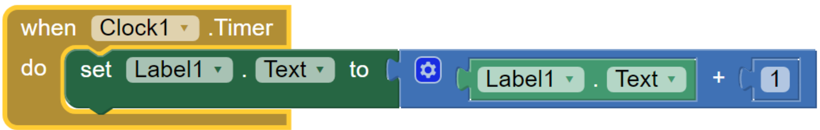
Step 2: Change the properties of the clock

○ The properties "TimerEnabled" specifies whether the Timer event should run, its value can be "True" or "False".
○ The properties "TimerInterval" specifies how long the clock will fire the Timer event.
○ TimerInterval:
  1. TimerInterval = 1000 → 1 second
  2. TimerInterval = 500 → 0.5 seconds
  3. TimerInterval = 3000 → 3 seconds
  4. If we want to perform a task 10 times in a second, the TimerInterval = 1000 / 10 = 100



Step 3: Add blocks to the clock

○ You can add blocks in the event "When Clock1.Timer do …".
○ For example, we add the value of Label1.Text by 1 every "TimerInterval" (1000 ms).



Notes

4.2 Homework

Task 1 (5 marks): - Adding the **countdown**[1] feature to the game.
In MIT App Inventor 2, finish the following steps:

Step 1: Open the project "BPST_2E01", which was built in last week.

Step 2: Make sure that you have finished all interfaces design (Designer Mode).

- Add a component "Sensors → Clock" to the screen "Game".

Step 3: Add blocks to do the following (Blocks Mode).

- To control the countdown with "Sensors → Clock".
- To display the time left with Label..

Step 4: Export your project to a file (.aia) and submit to the Schoology.

Task 2 (5 marks): - Keeping **program log**[2].
Please copy the programming code from your project. You should keep the log in a systematic way. Therefore, in the spaces provided below, please write down the blocks that you have used in different screens. An example in "Screen1" is shown below.

Screen "Screen1"

```
When Button1.Click
do  open another screen with start value    screenName    "Game"
                                            startValue      0
```

Please keep your program log in "Worksheet 04".

[1] Countdown 倒數
[2] Program Log 程式日誌

Chapter 5 Add images and background music in an App

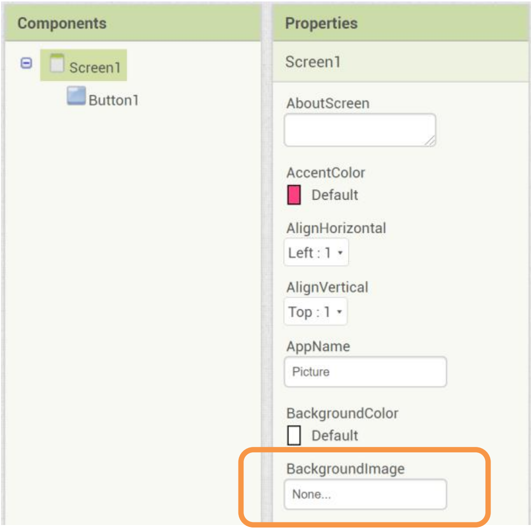
Key Words

Upload 上載	Source 來源	

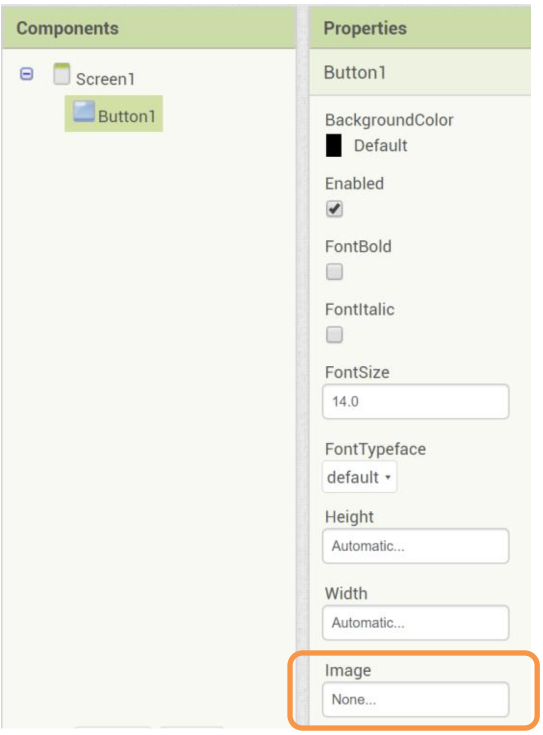
5.1 Adding images

Notes

- You can **upload**[1] background image to the screen "Screen1" by editing the property: "Screen1.BackgroundImage".



- You can change the image of the component "Button1" by editing the property: "Button1.Image".

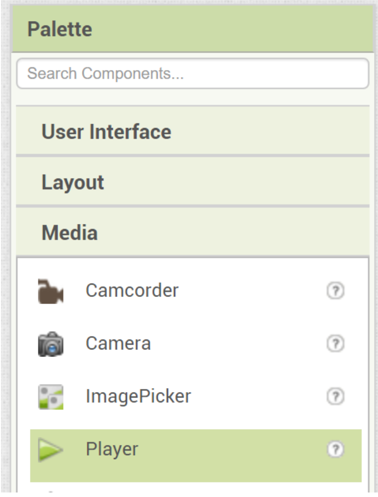


[1] Upload 上載

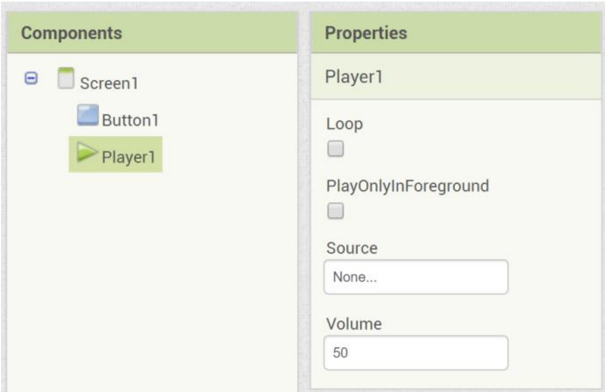
Notes

5.2 Adding background music

- You should add a component "Media → Player" to the screen.



- You can upload a MP3 music file to the component "Player1" by editing the property "Player1.Source".



- Properties of the component "Player":
  1. *Player1.Loop*
     If checked (value → True), the music will be played again after finish.

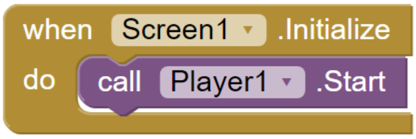
  2. *Player1.PlayOnlyInForeground*
     If checked (value → True), the music will stop when leaving the current screen.

  3. *Player1.Source*
     Upload or change the music file **source**[1].

  4. *Player1.Volume*
     Control the loudness of the music in terms of percentage (0 to 100)

[1] Source 來源

- Start playing the background music in Block mode.



5.3 Homework

<u>Task 1 (10 marks)</u>: - Adding the images and background music to the game.
In MIT App Inventor 2, finish the following steps:

Step 1: Open the project "BPST_2E01", which was built in last week.

Step 2: Make sure that you have added background image to every screen.
And change the image of every button

Step 3: Add blocks to do the following (Blocks Mode).

- To start and stop the background music.

Step 4: Export your project to a file (.aia) and submit to the Schoology.